Full-Stack Group Lab 1 Documentation

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1 State Management - Violet

1.1 Products

The products state is the central state of the program. Products is stored in App.jsx and contains all of the products for displaying.

1.2 Filtered-Products

The filteredProducts state stores the filtered products which is given out to the ProductTile components for displaying. Filtered products is changed when the filter inputs change or when the overall products change causing it to re-filter the products. The filtered products are what gets displayed to the user even if no filtering is currently active. Due to filteredProducts being used by the product tiles it is also stored in App.jsx where the product grid is at. Each of the filtered products gets passed as a prop to product tiles via a map.

1.3 Search Filter

Search filter stores the current text search query. When the input for the search input changes it causes the search filter to change which also causes a rerender of the product tiles to match the new query. The search filter contains the raw text of the search box and assessing the queries is done by the filterProducts function. Queries can use relational operators (>=, <=, >, <, =, ==) to check price or they can use '&' as a delimiter for multiple combined queries. This filter is stored in App.jsx.

1.4 Select Filter

The Select filter acts very similar to search filtering but provides a more accessible interface for filtering. Rather than using a text query to filter the user just clicks on an option in a dropdown box of categories for product. When the state of the dropdown changes the select filter does too triggering a rerender of the product tiles. This filter, like searchFilter, is stored in App.jsx.

2 CRUD Psuedocode - Brayden

3 Developer Log

3.1 Violet

3.1.1 Filtering

To implement the filtering I decided to copy the style I've used in the past few programs. I used both a category select option and a text search filter. When filtering

the products it applies both of these to the list of products and returns the filtered list. I also added alphabetical sorting like in my previous programs. I originally designed the filtering as its own component called FilterForm but realized near the end that this would not work very well because of how React state works. Instead I moved it up to App.jsx and redesigned the heirarchy. This wasn't a problem as I did this on my own time when the component heirarchy was still in its infancy. Going into this project we didn't have enough time to plan things out and the FilterForm is when I started to realize this. After setting everything up I cemented a component heirarchy diagram to guide further features now that we had a better idea of what our product will look like.

I wasn't happy that I had to move the logic back to App since I know it will make things very messy in the long run. With me moving it back to App it resulted in around 150 lines of code for something that would be far better as a component. There may be a way to do it as a component but it would be such a headache to set up that it would wind up making things more complex. This is definitely a pitfall of React. The increased complexity in writing dynamic code hurts how modular you can make the program. I do think with a database things would be a lot better since database langauges like SQL include filtering clauses and sorting clauses which would trivialize this. So in all honestly you can't blame React too much but I must say that state is a headache to deal with as compared to something like classes and properties you might be working with if you were using a basic templating engine such as Thymeleaf (less reactive but is also SSR)

3.2 Brayden

3.3 Jarrod

3.4 Isaiah